

Introduction

Juvenile open-angle glaucoma (JOAG) is a subset of childhood glaucoma that occurs after 3 years of age and with no globe enlargement. JOAG is diagnosed in patients presenting with glaucoma in the context of a normal appearing anterior chamber angle without any evidence of a secondary cause.¹ It is inherited in an autosomal dominant fashion with variability in penetrance and expressivity. It is strongly linked to a myocilin gene (MYOC) mutation found within the GLCA locus.² Moreover, cytochrome P450, family 1, subfamily B (CYP1B1) mutations have also been linked to JOAG in some populations.³

Given the young age at presentation, a significant proportion of JOAG patients are likely to require surgery at some point in their lifetime.⁴ Previous reports in the literature have looked at the outcomes of variable surgical interventions (i.e., trabeculotomy⁵, trabeculectomy⁶, ab interno trabeculectomy⁷, and viscocanalostomy⁸) in patients with JOAG. Trabeculectomy was shown to be effective in controlling the intraocular pressure (IOP) with a 73% cumulative probability of success at 3 years.⁶ However, this was associated with visually significant complications such as hypotony maculopathy and cataract formation. Deep sclerectomy (DS) is a non-penetrating glaucoma surgery that is considered as an alternative option in patients with open-angle glaucoma. Compared to trabeculectomy, evidence suggests that DS can produce similar success rates with fewer postoperative complications.⁹ To date, the literature lacks studies investigating the outcomes of DS in patients with JOAG. Therefore, our current study aims to report our experience with DS in a serial cohort of JOAG patients.

Methods

This study was approved by the local Institutional Review Board. We performed a retrospective chart review of patients who underwent DS for JOAG at a single tertiary care eye center. The indications for surgery were IOP uncontrolled on maximum medical therapy or intolerance to topical medications. The following criteria were set to identify JOAG patients: case coded as open-angle glaucoma (ICD 10 code: H40.1), age of onset was between 4 and 40 years¹, and no underlying cause of secondary glaucoma. Patients that did not complete a postoperative follow-up of at least six months following DS were excluded.

Preoperative information collected from the patient's medical records include age, sex, number of anti-glaucoma medications, previous glaucoma surgeries, visual acuity, IOP, lens status, cup-disc-ratio, and visual field results (if available).

All surgical interventions were performed by either a glaucoma consultant or a fellow credentialed to perform DS (surgical privilege is acquired after performing a minimum of 15 cases). All procedures were performed following a standardized surgical technique. First, the surgery was started by a fornix-based conjunctival periotomy. Mitomycin C (0.2 mg/ml) was then applied over bare sclera for 3 minutes. This was followed by the creation of a superficial scleral flap that was 5 × 5 mm in dimensions and approximately one-third of the scleral thickness in depth. Then, a deep scleral flap was dissected, leaving a shallow scleral bed (around 50-100 µm thick). Dissection of the deep flap was advanced towards the limbus until the roof of Schlemm's canal is reached and removed. Percolation of aqueous humor through the remaining trabeculo-Descemet's membrane was used as a confirmatory intraoperative sign of successful filtration.

After that, the deep scleral flap was excised, and the superficial scleral flap was closed with a nylon suture. There was no use of a space maintainer in the bed of the deep scleral flap. Finally, the conjunctiva was closed with a vicryl suture.

Postoperatively, all patients received topical steroids (prednisolone acetate 1%) on a tapering regimen over 8 weeks, topical antibiotics for two weeks, and topical pilocarpine 1% (to prevent formation of peripheral anterior synechiae and/or iris root incarceration) for one month. Postoperative follow-up visits were scheduled according to the following intervals: 1 day, 1 week, 1 month, 3 months, 6 months, 12 months, and then once every 3-6 months. Postoperative information collected in each visit include number of anti-glaucoma medications, visual acuity, IOP, surgical complications, and the need for further surgical intervention.

Success was defined as IOP within a range of 5-21 mmHg. Failure was defined as IOP measured out of the target range (5-21 mmHg) at any two consecutive visits after the first 3 months, an additional major glaucoma procedure is required, or if the patient progressed to no light perception.